

Cloud Migration Checklist

A provider-neutral checklist for moving on-premises workloads to Azure or AWS. Work top to bottom: assess what you have, design the network and identity, secure it before production, migrate in waves with a way back, validate that it works, and optimise cost and governance from the start.

ASSESS → DESIGN → SECURE → MIGRATE → VALIDATE → OPTIMISE

01 Discovery & dependency mapping

- ☐ Inventory servers, applications, databases and storage
- ☐ Inventory scheduled jobs, APIs, DNS records and certificates
- ☐ Inventory authentication paths and external integrations
- ☐ Run dependency analysis to reveal real communication flows
- ☐ Identify dependencies not visible from a server inventory
- ☐ Record owners and business criticality per workload

02 Network & connectivity

- ☐ Check for IP address range overlap between on-prem and cloud
- ☐ Design subnets, security groups and network ACLs
- ☐ Plan route tables and confirm routing end to end
- ☐ Confirm DNS resolution for internal and external names
- ☐ Review firewall rules to be recreated in the target
- ☐ Design on-prem connectivity: VPN, ExpressRoute or Direct Connect

03 Application dependencies

- ☐ Map each workload to its database servers
- ☐ Map directory and authentication dependencies
- ☐ Map APIs, file shares and third-party services it needs
- ☐ Map the other applications that depend on it
- ☐ Determine the correct order to move dependent components

04 Migration strategy (per workload)

- ☐ Rehost: lift and shift as-is
- ☐ Replatform: minor changes, e.g. a managed database
- ☐ Refactor: re-architect for the cloud model
- ☐ Repurchase: move to a SaaS equivalent
- ☐ Retire: switch off what is no longer used
- ☐ Retain: leave in place for now
- ☐ Decision driven by business requirement, not one policy for all

05 Identity & security

- ☐ Plan directory integration (Entra ID / IAM identity source)
- ☐ Define role-based access control and least privilege
- ☐ Use workload identities (managed identities / IAM roles)
- ☐ Design the privileged access model
- ☐ Set up secrets and key management (Key Vault / Secrets Manager, KMS)
- ☐ Use private endpoints / PrivateLink for service traffic
- ☐ Enforce encryption in transit and at rest
- ☐ Confirm all of the above is in place before production

06 Data & database migration

- ☐ Evaluate target compatibility for each data store
- ☐ Size the data volume and the time to move it
- ☐ Plan replication or synchronisation to shrink downtime
- ☐ Define backup and restore approach
- ☐ Agree RPO and RTO with the business
- ☐ Agree the acceptable downtime window
- ☐ Design critical database moves around business needs, not the tool

07 Performance & right-sizing

- ☐ Measure CPU, memory, disk IOPS and network over a real period
- ☐ Do not copy on-prem VM sizes into the cloud
- ☐ Select resources to match measured load, with room to scale
- ☐ Use provider sizing advisors to inform the choice
- ☐ Right-size at migration time, before inertia sets in

08 Cutover & rollback

- ☐ Define migration waves and their order
- ☐ Agree maintenance windows
- ☐ Set validation checkpoints that must pass to proceed
- ☐ Plan DNS changes and when they take effect
- ☐ Keep data synchronised up to the switch
- ☐ Document a rollback plan for a failed wave
- ☐ Confirm you can describe exactly how to put a workload back

09 **Observability after migration**

- ☐ Enable monitoring and logging (Azure Monitor / CloudWatch)
- ☐ Enable application-level insight and tracing
- ☐ Create alerts and dashboards before go-live
- ☐ Validate that backups restore, do not assume they do
- ☐ Confirm you can see, understand and recover from an incident

10 **Cost & governance**

- ☐ Apply consistent tagging for cost and ownership
- ☐ Apply policy guardrails (Azure Policy / SCPs and Config)
- ☐ Organise resources sensibly (subscriptions / accounts, groups)
- ☐ Set budgets and cost alerts
- ☐ Define the RBAC model for the environment
- ☐ Turn on active cost monitoring from day one

✓ **Definition of done**

- ☐ The application is secure
- ☐ The application is connected
- ☐ The application is performant
- ☐ The application is observable
- ☐ The application is recoverable
- ☐ The application is cost-effective
- ☐ Not just: the VM started